```
%(a)
x = linspace(0,4*pi,200);

%(b)
for i = 1:1:200
    y1 = sin(x(i))
end

%(c)
for i = 1:1:200
    y2 = cos(x(i)/2)
end

%(d)
figure;
y1 = sin(x);
plot(x,y1,'b','Linewidth',1.5);

%(e)
hold on;
y2 = cos(x/2);
plot(x,y2,'r--','Linewidth',1);

%(f)
title('sine & cosine function');
xlabel('angle');
ylabel('amplitude')

%(g)
legend('sine wave','cosine wave');

%(h)
grid on;
```

```
y1 =

     0


y1 =

    0.0631


y1 =

    0.1260


y1 =

    0.1883
```

```
y1 =

    0.2499

y1 =

    0.3105

y1 =

    0.3699

y1 =

    0.4278

y1 =

    0.4840

y1 =

    0.5382

y1 =

    0.5903

y1 =

    0.6401

y1 =

    0.6873

y1 =

    0.7318

y1 =
```

```
    0.7733

y1 =

    0.8118

y1 =

    0.8470

y1 =

    0.8789

y1 =

    0.9072

y1 =

    0.9320

y1 =

    0.9530

y1 =

    0.9702

y1 =

    0.9836

y1 =

    0.9930

y1 =

    0.9985

y1 =
```

```
    1.0000

y1 =

    0.9975

y1 =

    0.9910

y1 =

    0.9806

y1 =

    0.9663

y1 =

    0.9481

y1 =

    0.9261

y1 =

    0.9005

y1 =

    0.8712

y1 =

    0.8385

y1 =

    0.8025
```

```
y1 =

    0.7632

y1 =

    0.7209

y1 =

    0.6758

y1 =

    0.6279

y1 =

    0.5775

y1 =

    0.5249

y1 =

    0.4701

y1 =

    0.4135

y1 =

    0.3552

y1 =

    0.2955

y1 =

    0.2346
```

```
y1 =

    0.1728

y1 =

    0.1103

y1 =

    0.0473

y1 =

   -0.0158

y1 =

   -0.0789

y1 =

   -0.1416

y1 =

   -0.2038

y1 =

   -0.2652

y1 =

   -0.3255

y1 =

   -0.3845

y1 =

   -0.4420
```

```
y1 =

   -0.4977

y1 =

   -0.5515

y1 =

   -0.6030

y1 =

   -0.6521

y1 =

   -0.6987

y1 =

   -0.7424

y1 =

   -0.7832

y1 =

   -0.8209

y1 =

   -0.8553

y1 =

   -0.8863

y1 =
```

```
   -0.9138

y1 =

   -0.9376

y1 =

   -0.9577

y1 =

   -0.9739

y1 =

   -0.9863

y1 =

   -0.9947

y1 =

   -0.9992

y1 =

   -0.9997

y1 =

   -0.9962

y1 =

   -0.9888

y1 =

   -0.9774

y1 =
```

```
    -0.9621

y1 =

    -0.9429

y1 =

    -0.9201

y1 =

    -0.8935

y1 =

    -0.8634

y1 =

    -0.8298

y1 =

    -0.7930

y1 =

    -0.7529

y1 =

    -0.7099

y1 =

    -0.6640

y1 =

    -0.6155
```

```
y1 =

   -0.5646

y1 =

   -0.5113

y1 =

   -0.4561

y1 =

   -0.3990

y1 =

   -0.3404

y1 =

   -0.2804

y1 =

   -0.2192

y1 =

   -0.1572

y1 =

   -0.0946

y1 =

   -0.0316

y1 =

    0.0316
```

```
y1 =

    0.0946

y1 =

    0.1572

y1 =

    0.2192

y1 =

    0.2804

y1 =

    0.3404

y1 =

    0.3990

y1 =

    0.4561

y1 =

    0.5113

y1 =

    0.5646

y1 =

    0.6155

y1 =

    0.6640
```

```
y1 =

    0.7099

y1 =

    0.7529

y1 =

    0.7930

y1 =

    0.8298

y1 =

    0.8634

y1 =

    0.8935

y1 =

    0.9201

y1 =

    0.9429

y1 =

    0.9621

y1 =

    0.9774

y1 =
```

```
    0.9888

y1 =

    0.9962

y1 =

    0.9997

y1 =

    0.9992

y1 =

    0.9947

y1 =

    0.9863

y1 =

    0.9739

y1 =

    0.9577

y1 =

    0.9376

y1 =

    0.9138

y1 =

    0.8863

y1 =
```

```
    0.8553

y1 =

    0.8209

y1 =

    0.7832

y1 =

    0.7424

y1 =

    0.6987

y1 =

    0.6521

y1 =

    0.6030

y1 =

    0.5515

y1 =

    0.4977

y1 =

    0.4420

y1 =

    0.3845
```

```
y1 =

    0.3255

y1 =

    0.2652

y1 =

    0.2038

y1 =

    0.1416

y1 =

    0.0789

y1 =

    0.0158

y1 =

   -0.0473

y1 =

   -0.1103

y1 =

   -0.1728

y1 =

   -0.2346

y1 =

   -0.2955
```

```
y1 =

   -0.3552

y1 =

   -0.4135

y1 =

   -0.4701

y1 =

   -0.5249

y1 =

   -0.5775

y1 =

   -0.6279

y1 =

   -0.6758

y1 =

   -0.7209

y1 =

   -0.7632

y1 =

   -0.8025

y1 =

   -0.8385
```

```
y1 =

   -0.8712

y1 =

   -0.9005

y1 =

   -0.9261

y1 =

   -0.9481

y1 =

   -0.9663

y1 =

   -0.9806

y1 =

   -0.9910

y1 =

   -0.9975

y1 =

   -1.0000

y1 =

   -0.9985

y1 =
```

```
    -0.9930

y1 =

    -0.9836

y1 =

    -0.9702

y1 =

    -0.9530

y1 =

    -0.9320

y1 =

    -0.9072

y1 =

    -0.8789

y1 =

    -0.8470

y1 =

    -0.8118

y1 =

    -0.7733

y1 =

    -0.7318

y1 =
```

```
   -0.6873

y1 =

   -0.6401

y1 =

   -0.5903

y1 =

   -0.5382

y1 =

   -0.4840

y1 =

   -0.4278

y1 =

   -0.3699

y1 =

   -0.3105

y1 =

   -0.2499

y1 =

   -0.1883

y1 =

   -0.1260
```

```
y1 =

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y1 =

  -4.8986e-16

y2 =

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y2 =

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y2 =

    0.9980

y2 =

    0.9955

y2 =

    0.9920

y2 =

    0.9876

y2 =

    0.9821

y2 =

    0.9757

y2 =

    0.9683
```

```
y2 =

    0.9599

y2 =

    0.9506

y2 =

    0.9403

y2 =

    0.9291

y2 =

    0.9169

y2 =

    0.9039

y2 =

    0.8899

y2 =

    0.8751

y2 =

    0.8594

y2 =

    0.8428

y2 =

    0.8254
```

```
y2 =

    0.8072

y2 =

    0.7881

y2 =

    0.7683

y2 =

    0.7477

y2 =

    0.7264

y2 =

    0.7043

y2 =

    0.6815

y2 =

    0.6581

y2 =

    0.6340

y2 =

    0.6093

y2 =
```

```
    0.5839

y2 =

    0.5580

y2 =

    0.5316

y2 =

    0.5046

y2 =

    0.4770

y2 =

    0.4491

y2 =

    0.4206

y2 =

    0.3918

y2 =

    0.3625

y2 =

    0.3329

y2 =

    0.3030

y2 =
```

```
    0.2728

y2 =

    0.2423

y2 =

    0.2115

y2 =

    0.1806

y2 =

    0.1494

y2 =

    0.1181

y2 =

    0.0867

y2 =

    0.0552

y2 =

    0.0237

y2 =

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y2 =

   -0.0395
```

```
y2 =

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y2 =

   -0.1024

y2 =

   -0.1338

y2 =

   -0.1650

y2 =

   -0.1961

y2 =

   -0.2269

y2 =

   -0.2575

y2 =

   -0.2879

y2 =

   -0.3180

y2 =

   -0.3478

y2 =

   -0.3772
```

```
y2 =

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y2 =

   -0.4349

y2 =

   -0.4631

y2 =

   -0.4909

y2 =

   -0.5181

y2 =

   -0.5449

y2 =

   -0.5711

y2 =

   -0.5967

y2 =

   -0.6217

y2 =

   -0.6461

y2 =

   -0.6699
```

```
y2 =

   -0.6930

y2 =

   -0.7154

y2 =

   -0.7371

y2 =

   -0.7581

y2 =

   -0.7783

y2 =

   -0.7977

y2 =

   -0.8164

y2 =

   -0.8342

y2 =

   -0.8512

y2 =

   -0.8673

y2 =
```

```
   -0.8826

y2 =

   -0.8970

y2 =

   -0.9105

y2 =

   -0.9231

y2 =

   -0.9348

y2 =

   -0.9455

y2 =

   -0.9554

y2 =

   -0.9642

y2 =

   -0.9721

y2 =

   -0.9790

y2 =

   -0.9850

y2 =
```

```
    -0.9899

y2 =

    -0.9939

y2 =

    -0.9969

y2 =

    -0.9989

y2 =

    -0.9999

y2 =

    -0.9999

y2 =

    -0.9989

y2 =

    -0.9969

y2 =

    -0.9939

y2 =

    -0.9899

y2 =

    -0.9850
```

```
y2 =

   -0.9790

y2 =

   -0.9721

y2 =

   -0.9642

y2 =

   -0.9554

y2 =

   -0.9455

y2 =

   -0.9348

y2 =

   -0.9231

y2 =

   -0.9105

y2 =

   -0.8970

y2 =

   -0.8826

y2 =

   -0.8673
```

```
y2 =

   -0.8512

y2 =

   -0.8342

y2 =

   -0.8164

y2 =

   -0.7977

y2 =

   -0.7783

y2 =

   -0.7581

y2 =

   -0.7371

y2 =

   -0.7154

y2 =

   -0.6930

y2 =

   -0.6699

y2 =

   -0.6461
```

```
y2 =

   -0.6217

y2 =

   -0.5967

y2 =

   -0.5711

y2 =

   -0.5449

y2 =

   -0.5181

y2 =

   -0.4909

y2 =

   -0.4631

y2 =

   -0.4349

y2 =

   -0.4063

y2 =

   -0.3772

y2 =
```

```
   -0.3478

y2 =

   -0.3180

y2 =

   -0.2879

y2 =

   -0.2575

y2 =

   -0.2269

y2 =

   -0.1961

y2 =

   -0.1650

y2 =

   -0.1338

y2 =

   -0.1024

y2 =

   -0.0710

y2 =

   -0.0395

y2 =
```

```
   -0.0079

y2 =

    0.0237

y2 =

    0.0552

y2 =

    0.0867

y2 =

    0.1181

y2 =

    0.1494

y2 =

    0.1806

y2 =

    0.2115

y2 =

    0.2423

y2 =

    0.2728

y2 =

    0.3030
```

```
y2 =

    0.3329

y2 =

    0.3625

y2 =

    0.3918

y2 =

    0.4206

y2 =

    0.4491

y2 =

    0.4770

y2 =

    0.5046

y2 =

    0.5316

y2 =

    0.5580

y2 =

    0.5839

y2 =

    0.6093
```

```
y2 =

    0.6340

y2 =

    0.6581

y2 =

    0.6815

y2 =

    0.7043

y2 =

    0.7264

y2 =

    0.7477

y2 =

    0.7683

y2 =

    0.7881

y2 =

    0.8072

y2 =

    0.8254

y2 =

    0.8428
```

```
y2 =

    0.8594

y2 =

    0.8751

y2 =

    0.8899

y2 =

    0.9039

y2 =

    0.9169

y2 =

    0.9291

y2 =

    0.9403

y2 =

    0.9506

y2 =

    0.9599

y2 =

    0.9683

y2 =
```

```
    0.9757

y2 =

    0.9821

y2 =

    0.9876

y2 =

    0.9920

y2 =

    0.9955

y2 =

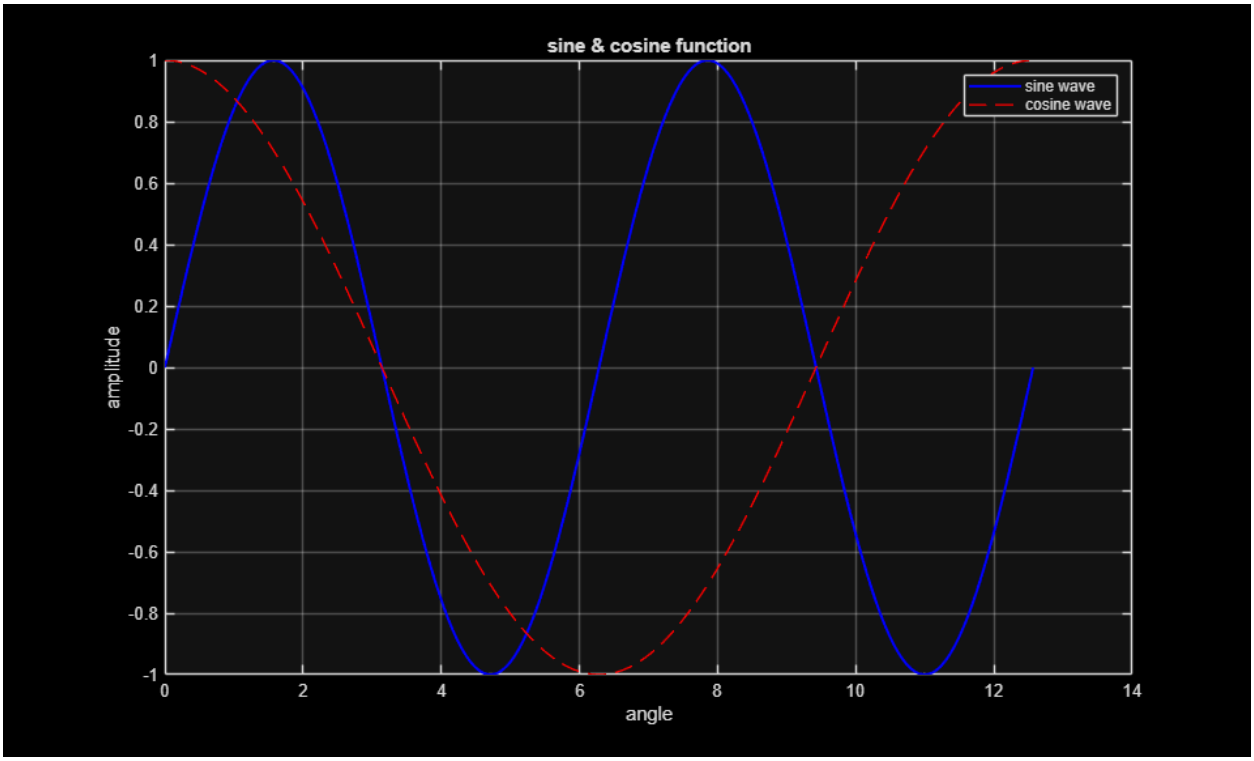
    0.9980

y2 =

    0.9995

y2 =

     1
```



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